

Sea Ice Deep Fusion

Workflow Overview

Combining satellite imagery with ICESat-2 laser altimetry to map Arctic sea ice: thick ice, thin ice, and open water

Research Seminar · Sea Ice Analysis

Metric	Value	Description
Test mIoU	0.8982	Mean intersection over union on held-out tile T03CWT
Pixel Accuracy	95.3%	Fraction of correctly classified pixels
Macro F1	0.9453	Unweighted average F1 across all three classes

Target Classes

Class	Label	Description
1	Thick Ice	Multi-year or first-year consolidated ice
2	Thin Ice / Melt Pond	Newly formed or flooded surface ice
3	Open Water	Ice-free ocean surface

3-class segmentation · tile-grouped train/test split · focal loss

1. Problem Statement and Data

1.1 Why Multimodal Fusion Is Required

- RGB imagery alone cannot reliably distinguish thin ice from thick ice; both appear nearly identical in visible wavelengths.
- Melt ponds introduce additional spectral confusion within the visible signal.
- Thin ice is rare in the dataset yet critically important for climate research.
- The test tile (T03CWT) represents a geographically distinct region with no spatial overlap with the training tiles, ensuring an honest generalisation test.

1.2 Proposed Approach

- Satellite RGB patches (128×128 px) provide spatial texture; a U-Net architecture processes this modality.
- ICESat-2 ATL03 height and roughness features provide altimetric context; an LSTM processes the along-track sequence.
- Feature sets from both branches are merged prior to the final classification layer.
- A ResNet-18 encoder initialised from ImageNet weights is used for the image branch.
- A Squeeze-and-Excitation (SE) block re-weights channels after the fusion step.

1.3 Data Sources

Source	Description
Satellite patches	128×128 px RGB crops. Training tiles: T02CNA, T02CNC. Test tile: T03CWT (held out entirely). Normalised
ICESat-2 ATL03	Along-track photon data at ~10 m spacing. Eight features per segment: h_cor_mean, h_diff, rel_height_mi
Label masks	RGB-coded segmentation maps. Red (255, 0, 0) = thick ice; Blue (0, 0, 255) = thin ice / melt pond; Green

1.4 Train / Validation / Test Split

Split	Tiles	Proportion	Notes
Train	T02CNA + T02CNC	~90%	Primary training data
Validation	T02CNA + T02CNC	~10%	Random held-out subset of train tiles
Test	T03CWT only	—	Completely unseen; geographically distinct from training region

Keeping T03CWT entirely out of training ensures that test metrics reflect genuine generalisation rather than spatial memorisation.

2. Model Architecture

The model consists of two parallel branches whose representations are concatenated and passed through a shared fusion head.

2.1 Image Branch (U-Net)

- Input: RGB patch of shape $B \times 3 \times 128 \times 128$.
- Encoder: ResNet-18 pretrained on ImageNet. Backbone weights are loaded and fine-tuned at the standard learning rate.
- Decoder: U-Net-style bilinear upsampling to restore full spatial resolution.
- Output: $B \times 16 \times 128 \times 128$ feature map.

2.2 CSV Branch (LSTM)

- Input: sliding window of 5 ATL03 segments, each with 8 features — shape $B \times 5 \times 8$.
- Sequence model: unidirectional single-layer LSTM, hidden size 96, tanh activation.
- The final hidden state h_T (shape $B \times 96$) is extracted.
- Two dense layers ($96 \rightarrow 16 \rightarrow 16$) with ELU activation and dropout project the hidden state to a 16-dimensional vector.
- The vector is tiled and expanded to match the spatial dimensions: $B \times 16 \times 128 \times 128$.

2.3 Fusion Head

- Concatenation of both branch outputs along the channel axis: $B \times 32 \times 128 \times 128$.
- Squeeze-and-Excitation (SE) block: global average pooling, two FC layers, sigmoid gating — re-weights channels adaptively.
- 3×3 convolution with BatchNorm, ReLU, and dropout.
- 1×1 convolution producing 3-class logits: $B \times 3 \times 128 \times 128$.
- Output: per-pixel class prediction $B \times 3 \times 128 \times 128$.

2.4 Training Configuration

Loss function	Focal loss; $\alpha = [0.05, 0.45, 0.60]$, $\gamma = 2.0$ (thin ice class up-weighted via $\alpha = 0.45$)
Optimiser	Adam
Learning rate	1×10^{-4} for newly initialised layers; 1×10^{-5} for pre-loaded LSTM weights
LR schedule	Cosine annealing
Precision	FP16 automatic mixed precision (AMP)
Early stopping	After 8 consecutive epochs without validation mIoU improvement
LSTM initialisation	Weights from the best LSTM sweep run (hidden = 96) are loaded into the CSV branch. In Fusion v4 the LSTM weights are frozen; in Deep Fusion v5 they are fine-tuned at $0.1 \times$ the base LR.

3. Training Procedure

Training follows a six-step pipeline. The LSTM branch is first optimised independently via a hyperparameter sweep, and the winning weights are then incorporated into the full fusion model.

Step 1: Match images to CSV rows

Each 128×128 patch is linked to an ATL03 row via tile identifier, beam identifier, and row_idx stored in the manifest file. Eight-feature arrays are built for every CSV segment.

Step 2: Normalise and cache CSV features

CSV features are z-scored using mean and standard deviation computed from training tiles only — test statistics never influence normalisation. Normalised arrays are saved as .npy files to avoid recomputation each epoch. Image normalisation uses ImageNet statistics and is applied on-the-fly. The training set receives random horizontal flips and 90° rotations.

Step 3: Identify optimal LSTM hyperparameters

Prior to constructing the fusion model, a sweep is conducted over hidden size {32, 64, 96}, learning rate, dropout, focal loss parameters (γ , α), and window length. Best configuration: hidden = 96, $lr \approx 1 \times 10^{-3}$, $\alpha = [0.05, 0.45, 0.60]$, $\gamma = 2.0$, seq_len = 5.

Step 4: Assemble the fusion model

A U-Net with a ResNet-18 encoder (ImageNet weights) produces a 16-channel feature map at full resolution. The sweep-winner LSTM weights are loaded into the CSV branch. The 96-dimensional hidden state is tiled to match the 128×128 spatial size before concatenation.

Step 5: Train with differentiated learning rates

Focal loss with $\alpha = 0.45$ on the thin ice class substantially reduces the tendency to ignore the rare class. Loaded (pretrained) layers train at $LR = 1 \times 10^{-4}$; freshly initialised layers train at $LR = 1 \times 10^{-3}$. Early stopping triggers after 8 epochs with no validation mIoU gain.

Step 6: Evaluate on held-out tile

The checkpoint with the highest validation mIoU (best.pt) is loaded and applied to T03CWT — a tile the model has never seen. Per-class IoU, precision, recall, F1, and macro averages are reported. The confusion matrix is row-normalised to percentages.

Training tiles: T02CNA + T02CNC. Test tile: T03CWT. No geographic overlap between the two sets — the reported numbers are honest out-of-distribution estimates.

4. Model Comparison

Five model variants are evaluated on the held-out test tile T03CWT. Results are summarised below.

4.1 Test mIoU — All Models

Model	mIoU	Pixel Acc.	IoU Thick	IoU Thin	IoU Water
U-Net (image only)	0.8704	94.29%	0.9299	0.7683	0.9130
LSTM (CSV only)	0.2420	72.54%	0.7254	0.0000	0.0005
Fusion Deep v1	0.8915	95.12%	0.9396	0.8015	0.9335
Fusion v4 ★ (best)	0.8982	95.31%	0.9407	0.8157	0.9382
Deep Fusion v5	0.8896	95.02%	—	—	—

★ Best overall: Fusion v4 — frozen sweep-winner LSTM + U-Net + SE fusion head.

4.2 Per-class IoU — Best Model (Fusion v4)

Class	IoU	Precision	Recall	F1
Thick Ice	0.9407	~0.97	~0.96	0.9695
Thin Ice	0.8157	~0.86	~0.93	0.8985
Water	0.9382	~0.99	~0.95	0.9681

5. Results Breakdown

Thin ice is the primary performance bottleneck. The following sections analyse each class and the key design decisions that contributed to the final results.

5.1 Confusion Matrix (Row-Normalised, Best Model)

	Pred: Thick Ice	Pred: Thin Ice	Pred: Water
Thick Ice	96.14%	3.86%	0.00%
Thin Ice	7.06%	92.78%	0.16%
Water	0.52%	5.01%	94.46%

5.2 Per-class Analysis

Thick ice — nearly clean: IoU 0.9407 / F1 0.9695. Only ~3% of thick ice pixels are mislabelled, making it the easiest class for the model.

Thin ice — primary bottleneck: IoU 0.8157 / F1 0.8985. Most confusion is with thick ice. The focal loss weight $\alpha = 0.45$ substantially mitigates the class imbalance.

Open water — texture discriminates well: IoU 0.9382 / F1 0.9681. Recall of 94.5% indicates that U-Net captures water surface texture effectively.

5.3 Key Findings

- **Fusion outperforms single-modality baselines.** LSTM alone achieves mIoU 0.24; U-Net alone achieves 0.87; fusion reaches 0.8982 — approximately 3 percentage points above the best single-modality model.
- **Pretrained LSTM weights provided free gains.** Loading the sweep-winner weights into the fusion model yielded +0.6 pp mIoU over random initialisation with no additional training time.
- **The SE block contributes consistent marginal improvements.** Re-weighting channels after concatenation improved thin ice IoU by approximately 1 percentage point while adding negligible parameters.
- **Tile-grouped splits are essential for honest evaluation.** Random splits would inflate scores due to spatial correlation across tiles.

6. Quantitative Results Summary

The table below consolidates all reported metrics across model iterations.

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★ Best overall (Fusion v4): frozen sweep-winner LSTM + U-Net + SE fusion head. Per-class IoU for Deep Fusion v5 was not recorded in the final report.

6.1 mIoU Progression Across Iterations

Iteration	Model	mIoU	Notes
1	U-Net only	0.870	Image modality baseline
2	LSTM only	0.242	CSV modality baseline; severely limited alone
3	Fusion Deep v1	0.891	First fusion attempt
4	Fusion v4 ★	0.898	Frozen LSTM weights; best result
5	Deep Fusion v5	0.890	Fine-tuned LSTM; slight regression

7. Conclusions and Future Work

7.1 Key Conclusions

- **C1 — Multimodal fusion is effective.** The best fusion model achieves mIoU 0.8982, compared to 0.87 for U-Net alone and 0.24 for LSTM alone.
- **C2 — Thin ice remains the weak point.** IoU 0.82 for thin ice, despite focal loss up-weighting ($\alpha = 0.45$). The class imbalance problem is partially addressed but not fully resolved.
- **C3 — LSTM pretraining provided cost-free improvement.** Loading sweep-winner weights yielded approximately +0.6 pp mIoU with no additional training budget.
- **C4 — Tile-grouped splits are critical for honest evaluation.** Spatially random splits would inflate scores due to cross-tile correlation.
- **C5 — The SE block is a small but consistent contributor.** The block adds almost no parameters but produces measurable improvements in thin ice IoU.

7.2 Future Work

- **F1 — Bi-directional LSTM.** A BiLSTM variant was tested standalone and showed improvement; integration into the fusion model is pending.
- **F2 — Test-time augmentation.** Averaging predictions over flipped and rotated inputs could reduce output variance without retraining.
- **F3 — Multi-date temporal stacking.** Incorporating imagery from multiple acquisition dates may improve performance near melt season transitions.
- **F4 — Attention mechanisms.** Replacing the fixed SE block with a full spatial attention layer may provide greater flexibility for edge cases.
- **F5 — Extended altimetry products.** Exploring ATL07 and ATL10 products would provide additional signal types beyond ATL03.

7.3 Technology Stack

Component	Details
Deep learning framework	PyTorch 2.x with FP16 AMP training
Segmentation library	segmentation-models-pytorch (smp) / U-Net
Image encoder	ResNet-18, pretrained on ImageNet
Altimetry data	ICESat-2 ATL03, along-track laser altimetry
Loss function	Focal loss for class-imbalance correction
Attention module	Squeeze-and-Excitation (SE) block for channel re-weighting

Best model (Fusion v4): mIoU 0.8982 · Pixel accuracy 95.3% · Macro F1 0.9453

Thick ice IoU 0.9407 · Thin ice IoU 0.8157 · Water IoU 0.9382
